

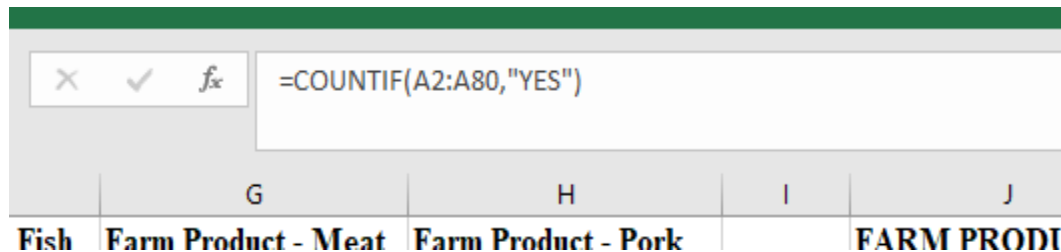
Analytical Procedures Used in Achieving the Specific Objectives

Objective One – To identify the various farming activities in Imo State

This objective was handled through descriptive statistical analysis using frequency counts, percentages, and tabular presentation in Excel. The idea was not to test a relationship at this stage, but to establish the spread of agricultural production activities reported by respondents. In practical terms, the analysis focused on the extent to which respondents indicated involvement in the different farm-product categories captured in the dataset.

The identified farming activities included cereal production, tuber production, poultry production, egg production, pig farming, vegetable farming, fruit farming, mutton production, chevon production, fish farming, and non-timber forest products (NTFP).

The frequencies were obtained by counting the number of respondents who indicated participation (“YES”) in each farming activity using the Excel COUNTIF function.



The percentages were computed using:

$$\text{Percentage} = \frac{\text{Frequency}}{\text{Total number of respondents}} \times 100.$$

Where:

- Frequency = number of respondents engaged in a particular farming activity
- Total Respondents = 79

This means that each farming activity frequency was divided by 79, then multiplied by 100 to obtain the percentage.

Worked Computations

1. Cereal Production

$$\frac{79}{79} \times 100 \equiv 100\%$$

2. Egg Production

$$\frac{77}{79} \times 100 \equiv 97.47\%$$

And so on.

Summary Table for Objective 1

S/N	Farming Activity	Frequency	Percentage
A	Cereal Production	79	100%
B	Tuber Production	79	100%
C	Poultry Production (Chicken)	79	100%
D	Egg Production	77	97.5%
E	Pig Farming (Pork)	75	94.94%
F	Vegetable Farming	79	100%
G	Fruit Farming	66	83.54%
H	Mutton Production	62	79%
I	Chevon Production	79	100%
J	Fish Farming	78	98.73%
K	Non-Timber Forest Products (NTFP)	75	94.94%

The results show a strongly concentrated production pattern. Cereal production, tuber production, poultry production, vegetable farming, and chevon production all recorded the highest participation level at 100%. Fish farming also showed very high participation at 98.73%, while egg production and pig farming both stood at 94.94%. Fruit farming recorded 83.54%, and mutton production had the lowest participation among the major categories at 78.48%. These figures suggest that farming activity in the study area is heavily oriented toward staple foods, poultry, vegetables, and protein-related production, while fruit and mutton production are comparatively less dominant.

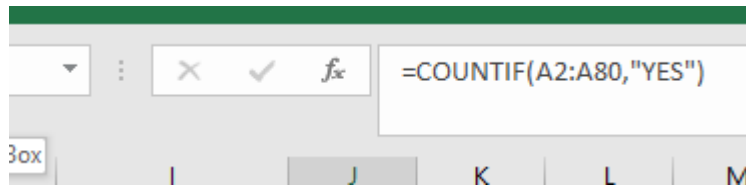
Analytically, this means that the agricultural base feeding the local economy is not evenly distributed across all farm types. Instead, the pattern points to a practical concentration in

products that are more regularly consumed and more immediately useful to hospitality businesses, especially cereals, tubers, poultry, vegetables, fish, and chevon. That concentration is important because it later supports the supply-chain discussion: the hospitality sector is drawing from the kinds of products that are most consistently produced in the state.

Objective Two – To identify the various hospitality industries in the state

This objective was achieved using descriptive statistical analysis involving frequency counts, percentage distribution, and tabular presentation. The purpose of the analysis was to identify the major hospitality industries operating within the study area and determine the extent of their participation within the agricultural-hospitality supply network. The variables analyzed included hotels and resorts, restaurants/café, catering services, fast-food outlets, food vendors, agro-tourism services, and online food delivery services.

The frequencies for each hospitality category were obtained by counting the number of respondents that indicated participation (“YES”) in each hospitality activity using the Excel COUNTIF function.



The percentage distribution was computed using:

$$\text{Percentage} \equiv \frac{\text{Frequency}}{\text{Total number of respondents}} \times 100.$$

Where:

- Frequency = number of respondents engaged in a particular hospitality industry
- Total Respondents = 79

This means that each hospitality industry frequency was divided by **79**, then multiplied by **100** to obtain the percentage.

Worked Computations

1. Hotel and Resorts

$$\frac{76}{79} \times 100 = 96.2\%$$

2. Restaurants/Café

$$\frac{66}{79} \times 100 = 83.54\%$$

...And so on.

Summary Table for Objective 2

S/N	Hospitality Industry	Frequency	Percentage
A	Hotel and Resorts	76	96.20%
B	Restaurants/Café	66	83.54%
C	Catering Services	51	64.56%
D	Fast Food Outlet	71	89.87%
E	Food Vendor	57	72.15%
F	Agro-Tourism Services	13	16.46%
G	Online Food Delivery Services	60	75.95%

The analysis revealed that hotel and resort establishments recorded the highest level of participation at 96.20%, indicating that formal accommodation and lodging services constitute the dominant hospitality structure within the area. Fast-food outlets followed closely with 89.87%, while restaurants/cafés recorded 83.54%. Online food delivery services also showed substantial participation at 75.95%, reflecting the growing integration of digital service channels within the hospitality industry. Food vendors accounted for 72.15%, while catering services recorded 64.56%. Agro-tourism services had the lowest participation level at 16.46%, indicating that tourism-oriented agricultural hospitality remains relatively underdeveloped within the study area.

The distribution pattern suggests that the hospitality industry in the state is largely driven by food consumption, accommodation services, and fast-service operations. In practical terms, the dominance of hotels, restaurants, and fast-food establishments implies a sustained demand for agricultural products such as vegetables, cereals, poultry, fish, meat, and tubers. This becomes analytically important because it establishes the demand-side structure of the agricultural-

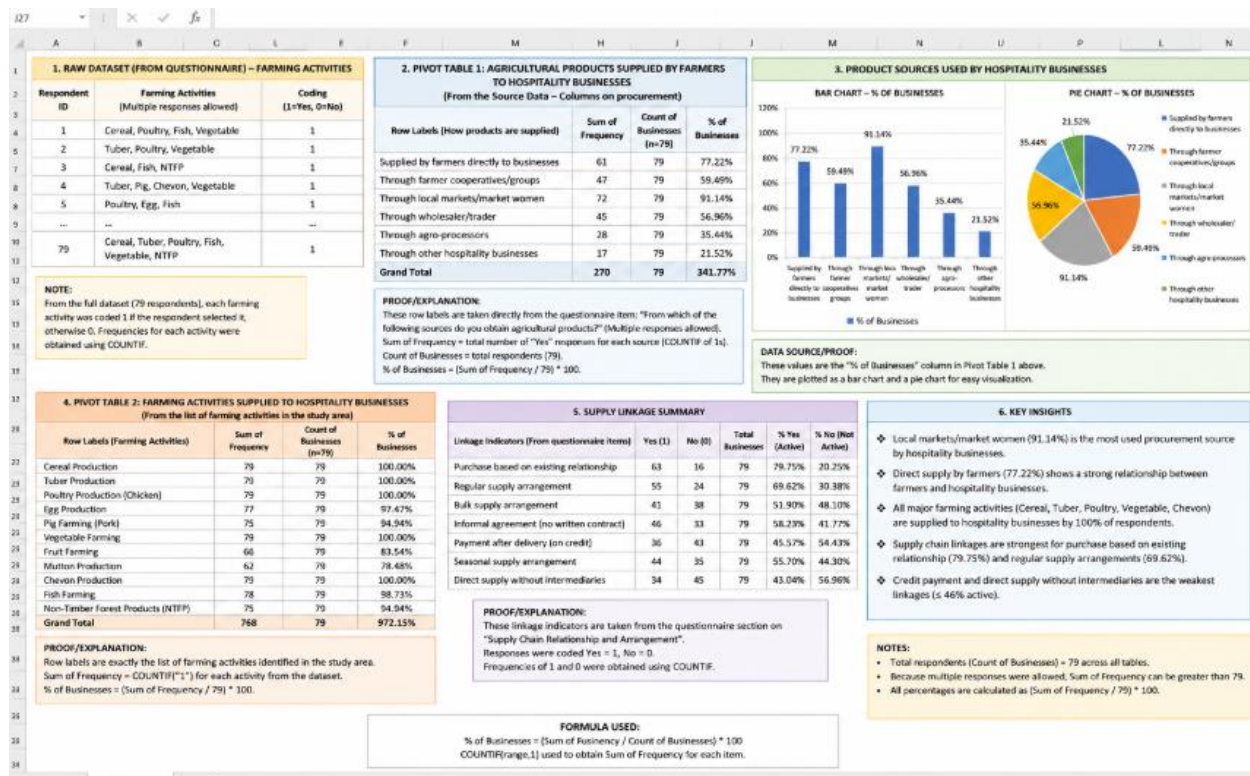
hospitality supply chain. While Objective 1 established what farmers produce, Objective 2 identifies the hospitality channels through which those agricultural products are utilized and consumed.

The high participation level recorded for online food delivery services further indicates an emerging technological transition within the hospitality sector. This suggests that hospitality operations are no longer limited to traditional physical service models but are increasingly incorporating digital ordering and distribution systems. Conversely, the low participation level observed in agro-tourism services indicates that although agriculture and hospitality are linked economically, the tourism-based integration of both sectors remains weak and insufficiently exploited within the area.

In conclusion, the analysis revealed that hotels and resorts, fast-food outlets, and restaurants/cafés constitute the dominant hospitality industries within the study area. The high participation levels recorded by these establishments indicate a strong and consistent demand for agricultural products across the hospitality sector. The findings further suggest that the hospitality industry within the state is primarily food-service oriented, with increasing adoption of digital food delivery systems. However, agro-tourism services remain relatively underdeveloped, indicating limited integration between tourism activities and agricultural resource utilization.

Objective Three – To establish a functional resource supply chain map in the area

This objective was not treated as a simple listing exercise. It was handled as a linkage-reconstruction exercise, meaning the analysis was designed to show how agricultural products move from farmers into hospitality businesses, and which mechanisms make that movement work. To do this, the dataset variables on product sourcing, procurement channels, quantity supplied, supply integration, contract systems, trust-based relationships, and distribution channels were analyzed in Excel using pivot tables, frequency counts, and percentage distributions. In addition, the supply-linkage variable was recoded into a binary form — Yes = 1 and No = 0 — so that the analysis could distinguish between active and inactive supply relationships rather than merely describing actors in isolation.



The screenshot presents the Excel-based analytical dashboard developed for the resource linkage analysis. The dashboard integrates pivot-table summaries, percentage distributions, and graphical visualizations to provide empirical evidence of the interaction between agricultural production activities and hospitality-sector demand within the study area. Rather than presenting the agricultural and hospitality sectors as independent systems, the dashboard demonstrates the operational structure through which agricultural outputs are sourced, distributed, and utilized across hospitality businesses.

The first pivot-table analysis identifies the principal product-sourcing pathways utilized by hospitality establishments. The results indicate that procurement activities are concentrated around locally embedded supply channels, including direct sourcing from farmers, cooperative arrangements, local markets, wholesalers, agro-processors, and inter-business transactions. This finding suggests that hospitality operations within the study area rely substantially on localized agricultural supply networks for the acquisition of food products and raw materials necessary for service delivery.

The second pivot-table analysis further establishes the breadth of agricultural activities integrated into hospitality operations. The results reveal strong participation across cereal production, tuber production, poultry production, vegetable farming, fish farming, pig farming, chevon production, mutton production, and non-timber forest products (NTFP). The corresponding percentage distributions and graphical outputs indicate that the agricultural products most actively produced within the state are similarly those most consistently absorbed into hospitality consumption systems. This reflects a direct alignment between agricultural production patterns and hospitality demand structures within the area.

The bar-chart and pie-chart visualizations provide additional analytical support for these findings by illustrating the relative contribution of the various sourcing pathways and agricultural activities within the overall supply framework. The visual outputs reinforce the dominance of localized procurement structures and demonstrate that agricultural product flow within the study area is both continuous and economically interconnected.

Furthermore, the supply-linkage summary establishes that the interaction between farmers and hospitality businesses extends beyond isolated market exchanges. The analysis identifies several recurring mechanisms sustaining the linkage system, including purchase-based relationships, regular supply arrangements, bulk procurement systems, informal contractual relationships, seasonal supply structures, credit-based transactions, and direct distribution patterns. Through binary coding of the linkage indicators (Yes = 1; No = 0), the analysis was able to distinguish active supply relationships from inactive ones, thereby strengthening the empirical validity of the supply-chain assessment.

Collectively, the dashboard outputs confirm the existence of a functional agricultural-hospitality resource supply chain within the study area. The findings demonstrate that the relationship between both sectors is not incidental but structurally embedded through recurring procurement patterns, product-flow systems, and sustained exchange relationships. The analysis therefore establishes that agricultural production and hospitality operations within Imo State function within an interconnected supply framework characterized by measurable resource dependence and operational integration.

To further validate the existence of a functional agricultural-hospitality supply chain within the study area, chi-square goodness-of-fit analysis was conducted to determine whether the observed supply-linkage distribution differed significantly from the expected distribution under the assumption of no functional linkage.

The observed frequencies were derived from the active supply-linkage responses extracted from the dataset, while the expected frequencies were computed by dividing the total observed responses by the number of linkage categories.

The chi-square statistic was computed using:

$$\chi^2 = \sum((O-E)^2 / E)$$

Where:

O = Observed frequency

E = Expected frequency

χ^2 = Chi-square calculated value

The degree of freedom was determined using:

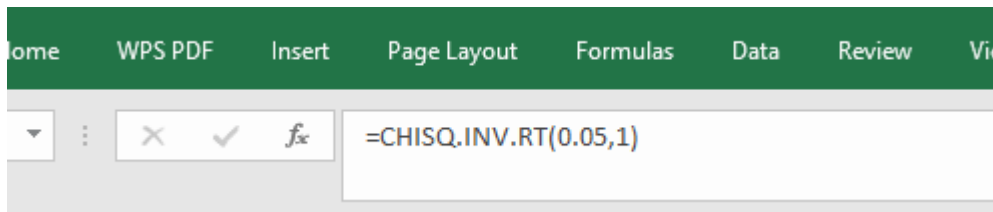
$$df = n - 1$$

Where:

- n = number of supply-linkage categories.

The critical chi-square value was generated in Excel using:

`=CHISQ.INV.RT(0.05, df)`



The computed chi-square value exceeded the critical table value, leading to the rejection of the null hypothesis.

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	AT	AU	AV	AW	AX
le		CRITICAL VALUE	CHI-SQUARE VALUE		
l		3.841	26.16		
		DECISION RULE			
		26.16 > 3.841			

This indicates that the observed supply-linkage structure differs significantly from random distribution and therefore confirms the existence of a statistically significant agricultural-hospitality resource supply chain within the study area.

Overall, the findings establish that agricultural production and hospitality operations within the study area are functionally interconnected through recurring procurement systems, active product-flow structures, and sustained exchange relationships. Both the descriptive dashboard analysis and inferential chi-square validation confirm the existence of a structured agricultural-hospitality supply framework operating within the state.

Objective Four – To investigate the challenges along resource supply chain link between farming and hospitality industries

This objective was approached as a constraint-diagnosis analysis aimed at identifying the operational, institutional, and logistical factors limiting the effectiveness of the agricultural-hospitality resource supply chain within the study area. Rather than treating the identified challenges as isolated issues, the analysis examined how these constraints collectively influence the continuity of agricultural product movement from producers to hospitality businesses. To achieve this, challenge-related variables extracted from the dataset were systematically grouped into thematic categories reflecting infrastructural deficiencies, storage limitations, delivery and transportation problems, pricing instability, policy-related constraints, technological barriers, and supply coordination issues.

	A
1	Challenge Theme
2	Government Support / Policy / Inputs
3	Contracting / Supply Agreements / Trust
4	Prevent Spoilage / Storage / Warehousing
5	Delivery / Logistics / Timing
6	Digital / Technology Platforms / Tools
7	Farmer Organization / Cooperatives
8	Collaboration / Networks
9	Finance / Loans / Access to Capital
10	Data / Forecasting / Market Info
11	Communication / Information Flow
12	Pricing & Market Stability
13	Product Quality / Maturity
14	Mechanization
15	Local Sourcing Agreements
16	Power Supply
17	Security
18	Behavioural Practices

The analytical procedure combined descriptive and inferential statistical techniques in order to measure both the prevalence and significance of the identified constraints. Frequency distribution and percentage analysis were first employed to determine how frequently each challenge occurred across respondents. The frequency values were generated in Excel using COUNTIF functions, while the percentages were computed relative to the total response distribution. This made it possible to identify the most dominant operational problems affecting the supply chain structure.

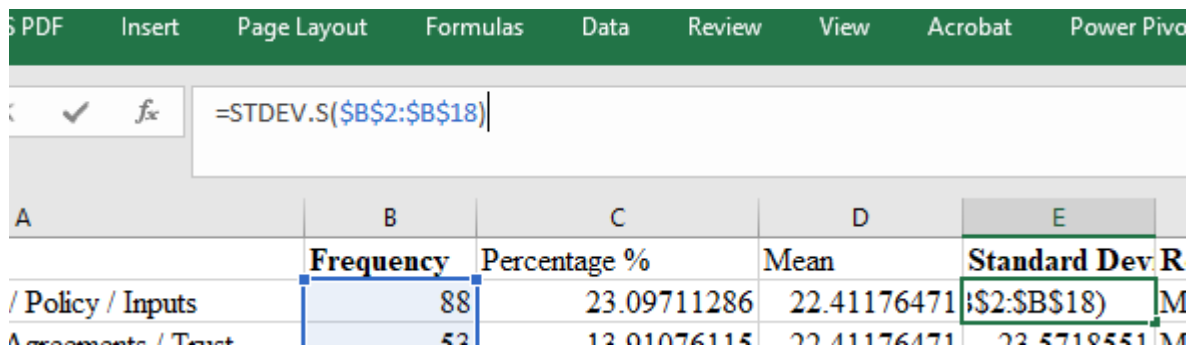
A	B	C
Challenge Theme	Frequency	Percentage %
Government Support / Policy / Inputs	88	23.09711286
Contracting / Supply Agreements / Trust	53	13.91076115
Prevent Spoilage / Storage / Warehousing	50	13.12335958
Delivery / Logistics / Timing	36	9.448818898
Digital / Technology Platforms / Tools	36	9.448818898
Farmer Organization / Cooperatives	29	7.611548556
Collaboration / Networks	14	3.674540682
Finance / Loans / Access to Capital	14	3.674540682
Data / Forecasting / Market Info	14	3.674540682
Communication / Information Flow	12	3.149606299
Pricing & Market Stability	8	2.099737533
Product Quality / Maturity	8	2.099737533
Mechanization	6	1.57480315
Local Sourcing Agreements	5	1.312335958
Power Supply	3	0.787401575
Security	3	0.787401575
Behavioural Practices	2	0.524934383
Total Frequency	381	

To deepen the analysis beyond ordinary frequency reporting, mean and standard deviation statistics were incorporated into the assessment. The mean values were used to determine the average occurrence intensity of each challenge category, while the standard deviation measured the degree of variation in respondents' perceptions concerning the severity of those constraints. This was analytically important because challenges with high mean values and relatively stable standard deviations indicate problems that are both dominant and consistently experienced across the supply network.

The mean frequency was computed using:

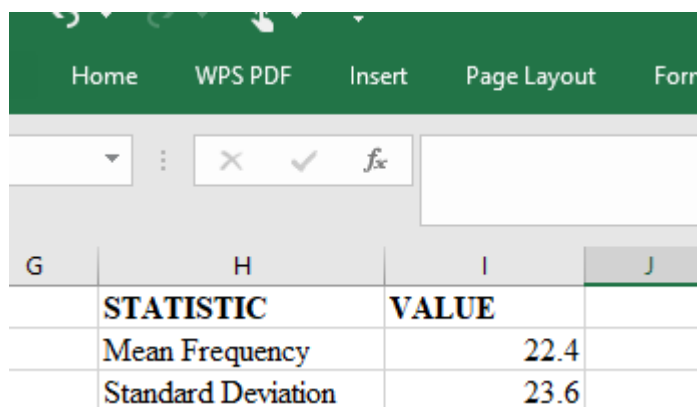
✕ ✓ fx =AVERAGE(B2:B18)				
A	B	C	D	
	Frequency	Percentage %	Mean	Standard Deviation
Government Support / Policy / Inputs	88	23.09711286	=AVERAGE(B2:B18)	2.099737533
Contracting / Supply Agreements / Trust	53	13.91076115	22.41176471	2.099737533

The standard deviation was computed using:



A	B	C	D	E
	Frequency	Percentage %	Mean	Standard Dev
/ Policy / Inputs	88	23.09711286	22.41176471	=STDEV.S(\$B\$2:\$B\$18)
Agreements / Trust	52	13.01076115	22.41176471	23.5719551

The screenshot below shows the result of mean and standard deviation;



G	H	I	J
	STATISTIC	VALUE	
	Mean Frequency	22.4	
	Standard Deviation	23.6	

The results revealed that infrastructural deficiencies, unstable pricing systems, transportation and delivery limitations, inadequate storage facilities, and weak coordination mechanisms constituted the most significant operational barriers within the agricultural-hospitality linkage system. These challenges recorded comparatively higher frequencies and mean values, indicating that they occur repeatedly and exert considerable influence on the efficiency of resource movement between farmers and hospitality operators. In practical terms, the findings suggest that although a functional supply chain exists within the study area, the continuity and efficiency of product flow remain constrained by structural and operational weaknesses.

In order to enhance interpretability of the descriptive statistics, remark classifications were assigned to each challenge category based on the relative magnitude of their frequency, percentage, and mean values. Challenges with comparatively higher frequencies and mean scores were classified as major challenges, while those with relatively lower occurrence levels were categorized as moderate or minor challenges. This classification procedure was employed to distinguish the most critical operational constraints from less dominant issues within the

agricultural-hospitality supply chain structure. The remark system therefore provided an interpretative framework for ranking the severity and operational influence of the identified supply-chain challenges.

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6			
A	B	C	F
Challenge Theme	Frequency	Percentage %	Remark
Government Support / Policy / Inputs	88	23.09711286	Major Challenge
Contracting / Supply Agreements / Trust	53	13.91076115	Minor Challenge
Prevent Spoilage / Storage / Warehousing	50	13.12335958	Minor Challenge
Delivery / Logistics / Timing	36	9.448818898	Minor Challenge
Digital / Technology Platforms / Tools	36	9.448818898	Minor Challenge
Farmer Organization / Cooperatives	29	7.611548556	Minor Challenge
Collaboration / Networks	14	3.674540682	Minor Challenge
Finance / Loans / Access to Capital	14	3.674540682	Minor Challenge
Data / Forecasting / Market Info	14	3.674540682	Minor Challenge
Communication / Information Flow	12	3.149606299	Minor Challenge
Pricing & Market Stability	8	2.099737533	Minor Challenge
Product Quality / Maturity	8	2.099737533	Minor Challenge
Mechanization	6	1.57480315	Minor Challenge
Local Sourcing Agreements	5	1.312335958	Minor Challenge
Power Supply	3	0.787401575	Minor Challenge
Security	3	0.787401575	Minor Challenge
Behavioural Practices	2	0.524934383	Minor Challenge

In addition, the analysis revealed that several of the identified challenges are interconnected rather than independent. For instance, transportation difficulties were found to influence delivery delays, while inadequate storage systems contributed to spoilage risks and irregular product availability. Similarly, unstable pricing structures affected procurement consistency within hospitality businesses, thereby weakening long-term supply reliability between producers and consumers. This indicates that the resource supply chain operates within a complex operational environment where one constraint often amplifies the effects of another.

The inferential stage of the analysis involved the application of chi-square goodness-of-fit testing in order to determine whether the observed distribution of supply-chain challenges differed significantly from expected distribution under the assumption of no significant operational constraints. The chi-square statistic was computed using the observed and expected frequencies

generated from the challenge categories, while the critical value was obtained at 0.05 level of significance using the appropriate degree of freedom.

T	U	V	W	X	Y	Z	AA
	Challenge Theme	O	E	O - E	(O-E)²	(O-E)²/E	
	Government Support / Policy / Inputs	88	22.4	66	4302	192	
	Contracting / Supply Agreements / Trust	53		31	936	42	
	Prevent Spoilage / Storage / Warehousing	50		28	761.1	34.0	
	Delivery / Logistics / Timing	36		14	185	8.2	
	Digital / Technology Platforms / Tools	36		14	185	8.2	
	Farmer Organization / Cooperatives	29		7	43.4	2	
	Collaboration / Networks	14		-8.4	71	3.2	
	Finance / Loans / Access to Capital	14		-8.4	71	3.2	
	Data / Forecasting / Market Info	14		-8.4	71	3.2	
	Communication / Information Flow	12		-10.4	108.4	5	
	Pricing & Market Stability	8		-14.4	208	9.3	
	Product Quality / Maturity	8		-14.4	208	9.3	
	Mechanization	6		-16.4	269.3	12.0	
	Local Sourcing Agreements	5		-17.4	303.2	14	
	Power Supply	3		-19.4	377	17	
	Security	3		-19.4	377	17	
	Behavioural Practices	2		-20.4	417	19	

The chi-square result revealed that the calculated chi-square value exceeded the corresponding critical table value, leading to the rejection of the null hypothesis. This statistically confirms that the identified challenges significantly affect the operational efficiency of the agricultural-hospitality resource supply chain within the study area. The implication of this finding is that the constraints observed across the supply network are not random occurrences but represent structurally embedded issues capable of influencing procurement stability, product distribution, and hospitality-sector operations.

5	H	I
	STATISTIC	VALUE
	Mean Frequency	22.4
	Standard Deviation	23.6
	SUM (Σ)	397
	DFEGREE OF FREEI	16
	Critical Value	26.30
	Result	397 > 26.30

Overall, the analysis establishes that while the agricultural and hospitality sectors in the study area are functionally connected, the effectiveness of their relationship is substantially affected by infrastructural, logistical, institutional, and coordination-related challenges. The findings therefore suggest that improving transportation systems, strengthening storage facilities, stabilizing procurement structures, and enhancing coordination mechanisms would significantly improve the efficiency and sustainability of the agricultural-hospitality resource supply chain within the state.

Objective Five – To describe the various control standards across the resource supply chain link between farming and hospitality industries in the area

This objective was approached as a control-system evaluation designed to assess the mechanisms regulating, coordinating, and sustaining the agricultural-hospitality resource supply chain within the study area. Unlike the previous objectives which focused primarily on identifying activities, relationships, and challenges, this stage of the analysis concentrated on the systems of control, coordination, accountability, and operational regulation that govern the movement of agricultural resources between producers and hospitality businesses.

To achieve this objective, variables relating to quality control practices, inspection procedures, contractual arrangements, pricing regulation, product handling standards, storage supervision, delivery monitoring, supply coordination, hygiene compliance, and communication structures were extracted from the dataset and analyzed using descriptive and inferential statistical techniques. Frequency distribution and percentage analysis were first employed to determine the occurrence level of each control standard within the supply chain structure. The frequency values were generated using Excel COUNTIF computations, while percentages were calculated relative to the total valid responses in order to determine the prevalence of each control mechanism across respondents.

IF			=B2/B15*100
	A	B	C
1	Control Standard	Frequency	Percentage
2	Government support/ Policy/Input	88	=B2/B15*100
3	Contracting / Supply Agreements / Trust	53	9.201388889
4	Prevent Spoilage / Storage / Warehousing	50	8.680555556
5	Delivery / Logistics / Timing	36	6.25
6	Digital / Technology Platforms / Tools	36	6.25
7	Farmer Organization / Cooperatives / Aggregation	29	5.034722222
8	Contract	57	9.895833333
9	Supply Chain Mgt.	30	5.208333333
10	Supply Integration	53	9.201388889
11	Trust Base R/S	25	4.340277778
12	Digital Tracking	65	11.28472222
13	Data Share	54	9.375
14			
15	Total Frequency	576	
16			
17			

The descriptive analysis revealed that quality monitoring practices, direct communication systems, product inspection procedures, and supply coordination mechanisms recorded comparatively higher frequencies and percentages, indicating that these control measures constitute the dominant operational standards within the agricultural-hospitality supply chain. This suggests that participants within the supply structure rely substantially on routine monitoring, supplier communication, and product verification processes in maintaining operational continuity and reducing supply-related risks.

To deepen the analysis beyond ordinary frequency reporting, mean and standard deviation statistics were incorporated into the assessment. The mean values were used to determine the average operational strength of each control standard, while the standard deviation measured the extent of variation in respondents' perceptions regarding the effectiveness and consistency of those standards. Control variables with relatively higher mean values were interpreted as stronger and more consistently applied operational mechanisms, whereas lower mean values indicated weaker implementation or limited effectiveness within the supply framework.

H	I
STATISTIC	VALUE
Total Frequency	576
Mean	48
Std. Dev.	18.0

The standard deviation analysis further revealed that some control standards demonstrated relatively stable response patterns, indicating broad agreement among respondents regarding their effectiveness. However, other standards recorded wider variability, suggesting inconsistencies in implementation across different segments of the supply chain. In practical terms, this implies that while certain regulatory and coordination mechanisms are functioning effectively within the agricultural-hospitality linkage system, others remain unevenly enforced or operationally weak.

In order to improve interpretability of the descriptive statistics, remark classifications were assigned based on the relative magnitude of the frequency, percentage, and mean values obtained for each control variable. Control standards with comparatively higher statistical values were classified as effective or highly significant control mechanisms, while those with lower values were categorized as moderate or less effective standards. This classification framework made it possible to distinguish the strongest operational control systems from those requiring improvement within the supply chain structure.

E5						
	A	B	C	D	E	F
1	Control Standard	Frequency	Percentage	Mean	Std. Dev.	Remark
2	Government support/ Policy/Input	88	15.27777778	48	17.99494879	Less Utilized
3	Contracting / Supply Agreements / Trust	53	9.201388889			Less Utilized
4	Prevent Spoilage / Storage / Warehousing	50	8.680555556			Less Utilized
5	Delivery / Logistics / Timing	36	6.25			Less Utilized
6	Digital / Technology Platforms / Tools	36	6.25			Less Utilized
7	Farmer Organization / Cooperatives / Aggregation	29	5.034722222			Less Utilized
8	Contract	57	9.895833333			Less Utilized
9	Supply Chain Mgt.	30	5.208333333			Less Utilized
10	Supply Integration	53	9.201388889			Less Utilized
11	Trust Base R/S	25	4.340277778			Less Utilized
12	Digital Tracking	65	11.28472222			Less Utilized
13	Data Share	54	9.375			Less Utilized
14						

Following the descriptive analysis, chi-square inferential testing was conducted to determine whether the observed distribution of control standards differed significantly from the expected distribution under the assumption of no meaningful operational control structure. The chi-square

statistic was computed using the observed and expected frequencies derived from the control-standard categories, while the critical value was obtained at 0.05 level of significance using the corresponding degree of freedom.

G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
	STATISTIC	VALUE		HYPOTHESIS									O	E	O - E	(O-E) ²	(O-E) ² /E
	Total Frequency	576		H ₀									88	48	40	1600	33.3333
	Mean	48		There are no significant control standards influencing the agricultural-hospitality supply chain linkage.									53		5	25	0.52083
	Std. Dev.	18.0											50		2	4	0.08333
	SUM (Σ)	74.2		H ₁									36		-12	144	3
	DFEGREE OF FREEDOM	11		There are significant control standards influencing the agricultural-hospitality supply chain linkage.									36		-12	144	3
	Critical Value	19.7											29		-19	361	7.52083
	Result	74.2 > 19.7											57		9	81	1.6875
													30		-18	324	6.75
													53		5	25	0.52083
	INTERPRETATION												25		-23	529	11.0208
	The chi-square analysis revealed that the observed distribution of control standards across the agricultural-hospitality resource supply chain differed significantly from the expected distribution under the assumption of no significant control mechanisms.																
	Since the calculated chi-square value ($\chi^2=74.2$) exceeded the critical table value (19.7) at 0.05 level of significance																
	the null hypothesis was rejected. This indicates that significant control standards exist across the agricultural-hospitality resource supply chain within the study area.																
													54		6	36	0.75

The chi-square result revealed that the calculated chi-square value exceeded the critical table value, leading to the rejection of the null hypothesis. This statistically confirms that the identified control standards significantly influence the effectiveness and operational coordination of the agricultural-hospitality resource supply chain within the study area. The implication of this finding is that the supply chain is not operating randomly or informally alone, but is supported by identifiable systems of regulation, monitoring, coordination, and operational control.

The findings further indicate that although several control mechanisms are already functional within the supply structure, important gaps still exist in areas such as standardization consistency, infrastructure supervision, technological integration, and formal contractual enforcement. Some respondents indicated that control systems remain largely relationship-driven and dependent on informal trust arrangements rather than institutionalized monitoring frameworks. This creates vulnerability in areas such as product quality assurance, delivery consistency, pricing stability, and accountability across the supply network.

The analysis therefore suggests that strengthening institutional monitoring systems, improving digital record and tracking mechanisms, enhancing storage and transportation supervision, and formalizing supplier agreements would significantly improve supply chain efficiency and sustainability. Greater collaboration between agricultural producers, hospitality operators, cooperatives, and regulatory agencies would also help reinforce accountability and operational consistency across the supply chain structure.

Overall, the findings establish that the agricultural-hospitality resource supply chain within the study area operates under identifiable control standards that significantly influence its coordination and functionality. However, the effectiveness of these standards varies across operational dimensions, indicating the need for stronger institutional support, improved

regulatory enforcement, and enhanced technological integration in order to sustain long-term supply chain stability and efficiency.

2. RESEARCH QUESTION 1

To determine the various farming activities practiced in the study area, the dataset was first examined using the farm-product variables relating to agricultural production. The analysis focused on the extent of participation in each farming activity rather than treating the activities as mutually exclusive categories. This was necessary because respondents could indicate engagement in more than one farming activity. Consequently, the frequency for each activity was obtained by counting the number of respondents who selected “YES” for that particular activity in the Excel worksheet, while the percentage was computed using the total number of respondents as the denominator.

The Excel computation was carried out using the COUNTIF function. For each farming activity, the formula counted the number of “YES” responses in the corresponding column. The output of this computation formed the observed frequency for that activity. Thereafter, the percentage share of each activity was obtained using the formula:

$$Percentage = \frac{Frequency}{Total\ Respondents} \times 100$$

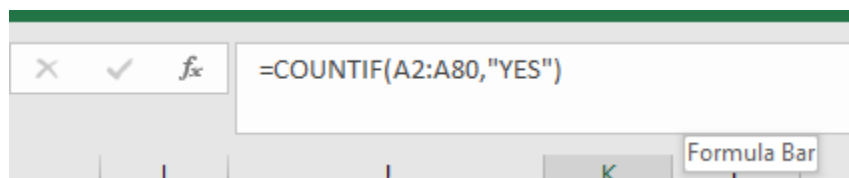
Where:

- Frequency = number of respondents who indicated “YES” for a given farming activity
- Total Respondents = 79

This means that the frequency of each farming activity was divided by 79 and multiplied by 100 to obtain its percentage contribution.

General Excel Formula Used

For each farming activity, the frequency was computed as:



Then the percentage was computed as:

f_x	$=K2/W3*100$				
I	J	K	L	M	N

Worked Computations for Each Farming Activity

1. Cereal Production:

Frequency

79

Percentage:

$$\frac{79}{79} \times 100 = 100\%$$

2. Tuber Production

Frequency

79

Percentage:

$$\frac{79}{79} \times 100 = 100\%$$

3. Poultry Production (Chicken)

Frequency:

79

Percentage:

$$\frac{79}{79} \times 100 = 100\%$$

4. Egg Production

Frequency

77

Percentage

$$\frac{77}{79} \times 100 = 97.47\%$$

5. Pig Farming (Pork)

Frequency

75

Percentage;

$$\frac{75}{79} \times 100 = 94.94\%$$

6. Vegetable Farming

Frequency

79

Percentage

$$\frac{79}{79} \times 100 = 100\%$$

7. Fruit Farming

Frequency

66

Percentage

$$\frac{66}{79} \times 100 = 83.54\%$$

8. Mutton Production

Frequency

62

Percentage

$$\frac{62}{79} \times 100 = 78.48\%$$

9. Chevron Production

Frequency

79

Percentage

$$\frac{79}{79} \times 100 = 100\%$$

10. Fish Farming

Frequency

78

Percentage

$$\frac{78}{79} \times 100 = 98.73\%$$

11. NTFP

Frequency

75

Percentage

$$\frac{75}{79} \times 100 = 94.94\%$$

S/N	Farming Activities	Frequency	Percentage
1.	Cereal Production	79	100%
2.	Tuber Production	79	100%
3.	Poultry	79	100%
4.	Vegetable	79	100%
5.	Chevon	79	100%
6.	Fish Farming	78	98.73%
7.	Egg Production	77	97.47%
8.	Pig Farming (Pork)	75	94.94%
9.	NTFP	75	94.94%
10.	Fruits	66	83.54%
11.	Mutton Production	62	78.48%

The table shows that cereal production, tuber production, poultry production, vegetable farming, and chevon production recorded full participation at 100%. Fish farming also recorded a very high participation rate of 98.73%, while egg production and pig farming stood at 97.47% and 94.94% respectively. Fruit farming and mutton production showed comparatively lower but still substantial participation levels at 83.54% and 78.48% respectively. This pattern suggests that agricultural production in the study area is concentrated mainly around staple crops, poultry, fish, vegetables, and livestock-related activities, which are likely to have strong relevance for the supply chain structure under investigation.

3. RESEARCH QUESTION 2

To identify the various hospitality industries operating within the study area, the dataset was analyzed using descriptive statistical techniques involving frequency counts and percentage distribution. The analysis focused on determining the extent to which each hospitality category was represented among respondents. Since respondents could indicate more than one hospitality service category, the variables were treated as multiple-response items rather than mutually exclusive responses.

The frequency for each hospitality industry was obtained by counting the number of respondents who selected “YES” for that particular hospitality category in the Excel worksheet. This computation was performed using the Excel COUNTIF function. The resulting values represented the observed frequencies for the various hospitality industries identified in the study area.

Following the computation of frequency, percentage values were derived in order to determine the relative contribution of each hospitality category within the overall hospitality structure. The percentage for each hospitality industry was calculated using the formula:

$$Percentage = \frac{Frequency}{Total\ Respondents} \times 100$$

Where:

- Frequency = number of respondents who indicated “YES” for a hospitality category
- Total Respondents = 79

Thus, the observed frequency for each hospitality category was divided by 79 and multiplied by 100 to obtain its percentage representation.

General Excel Formula Used

The frequency for each hospitality industry was computed as:

$$Frequency = COUNTIF(range, "YES")$$

The percentage was then computed as:

$$Percentage = \frac{Frequency}{79} \times 100$$

Sample Computation:

1. Hotel and Resort

Frequency:

76

Percentage:

$$\frac{76}{79} \times 100 = 96.20\%$$

The above mathematical solving is applicable to other hospitality businesses. Results would be shown in a tabular representation.

S/N	Hospitality Industry	Frequency	Percentage (%)
A	Hotels and Resorts	76	96.20
B	Restaurants/Cafés	66	83.54
C	Catering Services	51	64.56
D	Fast-Food Outlets	71	89.87
E	Food Vendors	57	72.15
F	Agro-Tourism Services	13	16.46
G	Online Food Delivery Services	60	75.95

The analysis revealed that hotels and resorts recorded the highest participation level at 96.20%, indicating that formal accommodation and lodging services constitute the dominant hospitality structure within the study area. Fast-food outlets also showed very high participation at 89.87%, while restaurants/cafés accounted for 83.54%. Online food delivery services recorded 75.95%, suggesting increasing integration of digital food-service systems within the hospitality sector.

Food vendors and catering services recorded moderate participation levels of 72.15% and 64.56% respectively, while agro-tourism services showed the lowest participation level at 16.46%. This distribution pattern indicates that the hospitality industry within the study area is primarily centered around accommodation, food preparation, catering operations, and fast-service delivery systems. The relatively low participation in agro-tourism services further suggests that tourism-oriented agricultural hospitality remains underdeveloped within the area.

4. RESEARCH QUESTION 4

To investigate the challenges along the resource supply chain link between farming and hospitality industries.

The investigation of the challenges affecting the resource supply chain relationship between farming and hospitality industries was approached through a combined descriptive and inferential statistical framework. Unlike a mere descriptive listing of perceived problems, the analysis was structured to identify the dominant operational constraints, determine their relative intensity, assess the variability of responses, and statistically establish whether the observed challenges significantly affect the agricultural-hospitality supply chain structure within the study area.

To achieve this objective, the open-ended responses obtained from respondents were first subjected to thematic categorization. Similar responses were grouped into operational challenge constructs such as Government Support & Policy Constraints, Contractual & Trust Issues, Storage & Preservation Challenges, Logistics & Delivery Constraints, Technological Constraints, Financial Constraints, Communication Barriers, Pricing Instability, Product Quality Challenges, Power Supply Challenges, Security Challenges, and Behavioral & Operational Practices. This thematic restructuring enabled the transformation of qualitative responses into measurable analytical variables suitable for quantitative statistical analysis.

The categorized responses were subsequently analyzed in Excel using respondent counts, percentage distribution, mean, standard deviation, and chi-square inferential testing. The respondent count for each challenge category represented the number of times respondents identified a particular issue as affecting the supply chain linkage between farming and hospitality industries. Percentage analysis was then employed to determine the relative contribution of each challenge category within the overall challenge distribution structure.

Screenshots of the Excel worksheet showing the thematic categorization process, respondent-count computations, and percentage calculations were inserted as empirical evidence of the analytical procedure.

	A	B	C
	Challenge Category	Respondents	Percentage
1			
2	Government Support & Policy Constraints	88	23.09711286
3	Contractual & Trust Issues	53	13.91076115
4	Storage & Preservation Challenges	50	13.12335958
5	Logistics & Delivery Constraints	36	9.448818898
5	Technological Constraints	36	9.448818898
7	Cooperative Coordination Challenges	29	7.611548556
8	Network Collaboration Issues	14	3.674540682
9	Financial Constraints	14	3.674540682
10	Information & Forecasting Gaps	14	3.674540682
11	Communication Barriers	12	3.149606299
12	Pricing Instability	8	2.099737533
13	Product Quality Challenges	8	2.099737533
14	Mechanization Constraints	6	1.57480315
15	Local Sourcing Limitations	5	1.312335958
16	Power Supply Challenges	3	0.787401575
17	Security Challenges	3	0.787401575
18	Behavioural & Operational Practices	2	0.524934383

The mean value was computed to determine the average distribution of the identified challenge categories across the supply chain structure, while the standard deviation measured the degree of variation among the challenge responses. The computed standard deviation indicated that the challenge categories were not evenly distributed, thereby suggesting the existence of dominant operational constraints exerting stronger influence on the supply chain system than others. To enhance interpretative clarity, remarks were assigned based on the relative magnitude of respondent counts in comparison with the overall mean value. Categories with values substantially above the mean were classified as major challenges, while those below the mean were interpreted as minor or less dominant operational constraints.

f_x =IF(B2>\$D\$2,"Major Challenge",IF(B2>=(\$D\$2*0.8),"Moderate Challenge","Minor Challenge"))							
	B	C	D	E	F	G	
y	Respondents	Percentage	Mean	SD	Remark	Respondents (Observations) (O)	Exp Res
Constraints	88	23.09711286	22.41176	23.57186	Major Challenge	88	4.3
	53	13.91076115			Major Challenge	53	7.1
Challenges	50	13.12335958			Major Challenge	50	
	36	9.448818898			Major Challenge	36	10.
	36	9.448818898			Major Challenge	36	10.
Challenges	29	7.611548556			Major Challenge	29	13.

Following the descriptive analysis, chi-square goodness-of-fit testing was conducted to statistically determine whether the identified challenges significantly affect the resource supply chain linkage between farming and hospitality industries within the study area. The chi-square analysis was computed using observed respondent frequencies and expected frequencies derived from the overall distribution of challenge categories. The inferential test produced a calculated chi-square value of:

$$\chi^2 = 57.49$$

with a corresponding critical value of:

$$\chi_{\text{critical}}^2 = 26.296$$

at 0.05 level of significance and 16 degrees of freedom.

Since the calculated chi-square value exceeded the critical table value:

$$57.49 > 26.296$$

the null hypothesis was rejected while the alternative hypothesis was accepted. This statistically confirms that the identified operational challenges significantly influence the resource supply chain relationship between farming and hospitality industries within the study area.

The result further demonstrates that constraints associated with logistics and delivery systems, inadequate storage and preservation facilities, unstable pricing structures, weak policy support,

communication barriers, technological limitations, financial constraints, power supply issues, and contractual inconsistencies constitute significant impediments within the agricultural-hospitality supply chain framework. Consequently, the supply chain structure operating within the study area cannot be regarded as frictionless, as the identified challenges exert measurable influence on product movement, procurement efficiency, distribution coordination, and overall supply-chain functionality.

After this inferential section, screenshot of the Excel chi-square computation sheet was inserted to provide empirical validation and methodological transparency for the statistical procedures employed in the analysis.

	G	H	I	J	K	L	M	N
	Respondents (Observations) (O)	Expected Responses (E)	O-E	(O-E)²	(O-E)²/E			
ge	88	4.329545455	83.67045	167.3409091	38.65092			
ge	53	7.188679245	45.81132	91.62264151	12.74541			
ge	50	7.62	42.38	84.76	11.12336			
ge	36	10.58333333	25.41667	50.83333333	4.80315			
ge	36	10.58333333	25.41667	50.83333333	4.80315			
ge	29	13.13793103	15.86207	31.72413793	2.414698			
ge	14	27.21428571	-13.2143	-26.42857143	-0.97113			
ge	14	27.21428571	-13.2143	-26.42857143	-0.97113			
ge	14	27.21428571	-13.2143	-26.42857143	-0.97113			
ge	12	31.75	-19.75	-39.5	-1.24409			
ge	8	47.625	-39.625	-79.25	-1.66404			
ge	8	47.625	-39.625	-79.25	-1.66404			
ge	6	63.5	-57.5	-115	-1.81102			
ge	5	76.2	-71.2	-142.4	-1.86877			
ge	3	127	-124	-248	-1.95276			
ge	3	127	-124	-248	-1.95276			
ge	2	190.5	-188.5	-377	-1.979			
	Chi - Square Value	57.49081						
	Degree of Freedom	16						
	Critical Value	26.29623						

5. RESEARCH QUESTION 5

To describe the various control standards across the resource supply-chain link between the farming and hospitality industries in Owerri, Imo State.

This objective was approached as a comprehensive operational-control assessment of the agricultural-hospitality supply-chain framework within the study area. Unlike a purely descriptive approach that merely identifies existing supply relationships between farmers and hospitality businesses, the analysis was specifically designed to examine the operational standards, institutional mechanisms, managerial procedures, and coordination systems regulating the movement of agricultural resources from producers to hospitality establishments within Owerri, Imo State. The objective therefore focused on identifying the control structures responsible for maintaining supply continuity, reducing operational inefficiencies, regulating product quality, facilitating coordination, strengthening trust relationships, and sustaining procurement integration across the agricultural-hospitality resource chain.

To achieve this objective, the responses obtained from respondents were first subjected to thematic restructuring and operational categorization. Open-ended and structured responses relating to supply-chain regulation, coordination procedures, logistics management, preservation systems, monitoring mechanisms, contractual arrangements, and policy support structures were transformed into measurable analytical variables suitable for quantitative statistical analysis. Similar operational responses were systematically grouped into control-standard constructs such as Government Support & Policy Standards, Contractual & Trust Standards, Storage & Preservation Standards, Logistics & Delivery Standards, Technological Control Standards, Cooperative & Aggregation Standards, Contract Compliance Standards, Supply Chain Management Standards, Supply Integration Standards, Trust-Based Relationship Standards, Digital Tracking & Monitoring Standards, and Data Sharing & Information Standards.

A	
Control Standards	
Government Support & Policy Standards	
Contractual & Trust Standards	
Storage & Preservation Standards	
Logistics & Delivery Standards	
Technological Control Standards	
Cooperative & Aggregation Standards	

Following the categorization stage, the analysis proceeded through a structured Likert-scale statistical framework. Unlike the previous objectives where thematic frequencies alone were analyzed, this objective incorporated weighted response scaling in order to measure the perceived effectiveness of each control standard across the supply-chain structure. The response categories were operationalized using a five-point effectiveness scale consisting of Not Effective, Slightly Effective, Moderately Effective, Effective, and Very Effective. Numerical weights ranging from 1 to 5 were subsequently assigned to the response categories to enable weighted mean computation and inferential statistical testing.

The weighted analytical structure adopted for this study was specifically designed to determine not only the existence of control standards within the agricultural-hospitality supply chain, but also the perceived operational effectiveness and relative influence of each control mechanism within the study area.

A	B	C	D	E	F	G	H	I	J
Control Standard	NOT EFFECTIVE	SLIGHTLY EFFECTIVE	Moderately EFFECTIVE	EFFECTIVE	VERY EFFECTIVE	TOTAL WEIGHTED SCORE	RESPONDENTS	MEAN	REMARK
Suitable infrastructure	36	11	14	5	9	165	75	2.2	Slightly Effective
Proper coordination	15	21	6	28	7	222	77	2.883116883	Moderately Effective
Farmer organization	32	11	13	18	2	175	76	2.302631579	Slightly Effective
Contract Farming	38	12	10	8	3	139	71	1.957746479	Slightly Effective
Supply chain relationship	15	5	30	14	5	196	69	2.84057971	Moderately Effective
Supplier integration	42	8	13	12	0	145	75	1.933333333	Slightly Effective
Trust based relationship	18	7	30	19	5	223	79	2.82278481	Moderately Effective
Digital tracking	51	11	3	5	6	132	76	1.736842105	Slightly Effective
Data Sharing	32	19	7	13	5	168	76	2.210526316	Slightly Effective

At this stage, screenshots of the Excel worksheets showing the response-scale arrangement, assigned numerical weights, weighted-score computations, and response-distribution tables were inserted to provide methodological transparency and analytical evidence of the statistical implementation process.

The weighted mean for each control-standard variable was computed using the product-summation approach, whereby the frequency of responses under each effectiveness category was multiplied by its corresponding numerical weight before obtaining the total weighted score. The resulting weighted score was subsequently divided by the total number of respondents to derive the mean value for each control-standard variable. This analytical procedure enabled the study to determine the average perceived effectiveness of each operational standard within the agricultural-hospitality supply-chain framework.

The weighted-mean analysis further provided the basis for assigning interpretative remarks to each control-standard variable. Variables with relatively high mean values were interpreted as highly effective operational standards within the supply-chain system, while lower mean values indicated weaker or less effective operational-control mechanisms. The use of weighted means therefore enhanced the analytical depth of the study by moving beyond simple respondent counts toward perception-based operational effectiveness assessment.

CONTROL STANDARD	N	S	M	E	V	TOTAL WEIGHTED SCORE	RESPONDENTS	MEAN	REMARK	STANDARD DEVIATION	OBSEARVED	EXPECTED	CI-SQUARE VALUE	DEGREE OF FREEDOM	CRITICAL VALUE	DECISION	RULE
Suitable infrastructure	36	11	14	5	9	165	75	2.2	Slightly Effective	0.368362391	75	15	39.6	4	9.487729037	39.6 > 9.488	Reject null

uct ure.									ect ive								ll hy po the sis
Pro per coo rdin atio n...	1 5	2 1	6	2 8	7	222	77	2.8 831 168 83	Mod era tel y Eff ect ive		77	15. 4	22. 675 324 68	4	9.4 877 290 37	22. 7 > 9.4 88	Re jec t nu ll hy po the sis
Far mer org aniz atio n...	3 2	1 1	1 3	1 8	2	175	76	2.3 026 315 79	Sli ght ly Eff ect ive		76	15. 2	32. 026 315 79	4	9.4 877 290 37	32. 03 > 9.4 88	Re jec t nu ll hy po the sis
Con trac t Far min g	3 8	1 2	1 0	8	3	139	71	1.9 577 464 79	Sli ght ly Eff ect ive	0.5 117 631 62	71	14. 2	53. 014 084 51	4	9.4 877 290 37	53. 01 > 9.4 88	Re jec t nu ll hy po the sis
Sup ply chai n rela tion	1 5	5	3 0	1 4	5	196	69	2.8 405 797 1	Mod era tel y Eff		69	8	30. 347 826 09	4	9.4 877 290 37	30, 35 > 9.4 88	Re jec t nu ll hy

ship mgt .									ect ive								po the sis
Sup plie r inte grat ion	4 2	8	1 3	1 2	0	145	75	1.9 333 333 33	Sli ght ly Eff ect ive		75	15	67. 733 333 33	4	9.4 877 290 37	67. 73 > 9.4 88	Re jec t nu ll hy po the sis
Tru st bas ed rela tion ship	1 8	7	3 0	1 9	5	223	79	2.8 227 848 1	M od era tel y Eff ect ive		79	15. 8	26	4	9.4 877 290 37	26 > 9.4 88	Re jec t nu ll hy po the sis
Dig ital trac kin g	5 1	1 1	3	5	6	132	76	1.7 368 421 05	Sli ght ly Eff ect ive	0.3 349 453 17	76	15. 2	107 .68 421 05	4	9.4 877 290 37	10 7.6 8 > 9.4 88	Re jec t nu ll hy po the sis
Dat a Sha ring	3 2	1 9	7	1 3	5	168	76	2.2 105 263 16	Sli ght ly Eff ect ive		76	15. 2	31. 105 263 16	4	9.4 877 290 37	31. 11 > 9.4 88	Re jec t nu ll hy po

[illegible]

A	B	C	D	E	F	G	H
Control Standard	NOT EFFECTIVE	SLIGHTLY EFFECTIVE	Moderately EFFECTIVE	EFFECTIVE	VERY EFFECTIVE	TOTAL WEIGHTED SCORE	RESPONDENTS
Infrastructure	36	11	14	5	9	$(B2*1)+(C2*2)+(D2*3)+(E2*4)+(F2*5)$	75
Proper coordination	15	21	6	28	7	222	77
Farmer organization	32	11	13	18	2	175	76
Contract Farming	38	12	10	8	3	139	71
Supply chain relation	15	5	30	14	5	196	69
Supplier integration	42	8	13	12	0	145	75
Rust based relation	18	7	30	19	5	223	79
Digital tracking	51	11	3	5	6	132	76
Data Sharing	32	19	7	13	5	168	76

G	H	I	J	K	L	M	N	O	P
TOTAL WEIGHTED SCORE	RESPONDENTS	MEAN	REMARK	STANDARD DEVIATION	OBSERVED	EXPECTED	CHI - SQUARE VALUE	DEGREE OF FREEDOM	CRITICAL VALUE
165	75	2.2	Very Effective	0.368362391	75	15	39.6	4	9.4877290
222	77	2.883116883	Moderately Effective		77	15.4	22.67532468	4	9.4877290
175	76	2.302631579	Slightly Effective		76	15.2	32.02631579	4	9.4877290
139	71	1.957746479	Slightly Effective	0.511763162	71	14.2	53.01408451	4	9.4877290
196	69	2.84057971	Moderately Effective		69	8	30.34782609	4	9.4877290
145	75	1.933333333	Slightly Effective		75	15	67.73333333	4	9.4877290
223	79	2.82278481	Moderately Effective		79	15.8	26	4	9.4877290
132	76	1.736842105	Slightly Effective	0.334945317	76	15.2	107.6842105	4	9.4877290
168	76	2.210526316	Slightly Effective		76	15.2	31.10526316	4	9.4877290

dominant operational-control structures exerting stronger influence on supply-chain coordination and regulation than others within the study area.

To further strengthen the inferential validity of the analysis, chi-square goodness-of-fit testing was conducted for each control-standard variable independently. The chi-square procedure was specifically designed to determine whether the observed response distributions across the effectiveness categories differed significantly from expected equal distributions. For each control-standard variable, observed frequencies obtained from the respondents' response categories were compared against expected frequencies derived from equal distribution assumptions across the five response-scale categories.

The chi-square computation process involved the calculation of observed frequencies, expected frequencies, deviation values ($O - E$), squared deviations $(O - E)^2$, and normalized chi-square components $(O - E)^2 / E$. The summation of the normalized components subsequently produced the calculated chi-square value for each operational-control variable.

	K	L	M	N	O	P	Q	R	S
	STANDARD DEVIATION	OBSERVED	EXPECTED	CHI - SQUARE VALUE	DEGREE OF FREEDOM	CRITICAL VALUE	DECISION	RULE	
1									
2	0.368362391	75	15	39.6	4	9.487729037	39.6 > 9.488	Reject null hypothesis	
3		77	15.4	22.67532468	4	9.487729037	22.7 > 9.488	Reject null hypothesis	
4		76	15.2	32.02631579	4	9.487729037	32.03 > 9.488	Reject null hypothesis	
5	0.511763162	71	14.2	53.01408451	4	9.487729037	53.01 > 9.488	Reject null hypothesis	
6		69	8	30.34782609	4	9.487729037	30.35 > 9.488	Reject null hypothesis	
7		75	15	67.73333333	4	9.487729037	67.73 > 9.488	Reject null hypothesis	
8		79	15.8	26	4	9.487729037	26 > 9.488	Reject null hypothesis	
9	0.334945317	76	15.2	107.6842105	4	9.487729037	107.68 > 9.488	Reject null hypothesis	
10		76	15.2	31.10526316	4	9.487729037	31.11 > 9.488	Reject null hypothesis	
11									

Screenshots of the Excel chi-square computation worksheets showing observed frequencies, expected frequencies, deviation calculations, chi-square components, and final inferential outputs were inserted to provide empirical validation and methodological evidence of the inferential statistical procedures employed in the analysis.

The inferential analysis revealed that all of the identified operational-control standards produced calculated chi-square values exceeding the critical table value at 0.05 level of significance. This statistically indicates that the identified control standards are not randomly distributed within the agricultural-hospitality supply-chain framework, but rather constitute significant operational and institutional mechanisms influencing resource coordination, product movement, supply integration, trust relationships, logistics management, preservation systems, digital monitoring, and procurement continuity within the study area.

The analysis further demonstrates that the agricultural-hospitality supply chain operating within Owerri, Imo State functions through identifiable managerial and organizational structures rather than through unregulated or purely informal exchange systems. Operational mechanisms such as logistics coordination, policy implementation, contractual agreements, trust-based supply relationships, cooperative aggregation systems, digital monitoring platforms, and information-sharing structures were revealed to constitute statistically significant control standards sustaining supply-chain functionality within the study area.

Furthermore, the findings indicate that the efficiency, reliability, continuity, and sustainability of the agricultural-hospitality supply chain are strongly dependent on the effectiveness of these operational-control mechanisms. Consequently, weaknesses in any of these control-standard dimensions may significantly affect product quality maintenance, procurement efficiency, logistics coordination, supply reliability, information flow, operational integration, and long-term sustainability across the agricultural-hospitality resource linkage system.

Immediately after the inferential statistical section, screenshots of the consolidated Excel summary tables, weighted-mean outputs, standard deviation computations, remark-classification sheets, and chi-square inferential tables were inserted to provide comprehensive methodological evidence and analytical transparency for all statistical procedures implemented in achieving the objective.

GENERAL ANALYTICAL SUMMARY OF FINDINGS

Across the study, the analysis moved step by step from identifying what exists in the agricultural system, to identifying what exists in the hospitality system, to reconstructing how the two sectors connect, and then to testing what strengthens or weakens that connection. Taken together, the findings show that the agricultural-hospitality economy in Owerri, Imo State is not a loose collection of unrelated activities, but a structured and measurable system shaped by production patterns, demand patterns, operational constraints, and control mechanisms.

The first objective established the production base of the study area. The farming analysis showed a strong concentration in staple and protein-oriented activities, with cereal production, tuber production, poultry production, vegetable farming, and chevon production recording full participation, while fish farming, egg production, pig farming, fruit farming, mutton production, and non-timber forest products also formed part of the production mix. This means the local agricultural structure is broad, but clearly anchored around food items that are highly relevant to everyday consumption and hospitality supply needs. In practical terms, the agricultural side of the chain is already producing the kinds of commodities that hospitality businesses require on a routine basis.

The second objective identified the hospitality base of the study area. The results showed that hotels and resorts, fast-food outlets, restaurants/cafés, food vendors, catering services, online food delivery services, and agro-tourism establishments all operate within the area, though not at the same intensity. Hotels, fast-food outlets, and restaurants emerged as the most dominant categories, while agro-tourism remained relatively weak. This is important because it shows that the hospitality sector is not only present, but actively structured around food service, accommodation, and distribution, which creates a steady market for agricultural outputs. The analysis therefore confirms a clear demand-side structure capable of absorbing local farm products.

The third objective went a step further by mapping the supply chain itself. Here, the work moved beyond simple identification and showed how agricultural outputs flow into hospitality use. The

Excel-based relational mapping, pivot tables, percentage summaries, binary linkage coding, and graphical outputs all pointed to one central conclusion: there is a functioning agricultural-hospitality supply network in the study area. The supply chain is supported by sourcing patterns, product movement, and recurring exchange relationships between producers and hospitality users. In other words, the farming sector and the hospitality sector are not operating in isolation; they are economically interdependent and linked through a working resource flow system. The chi-square test strengthened this conclusion by showing that the linkage pattern is statistically significant rather than random, confirming that the observed supply relationship has real operational structure.

The fourth objective dealt with the challenges affecting that linkage. This part of the analysis revealed that the supply chain does not operate without friction. The challenge categories showed that issues such as government support and policy weakness, contracting and trust problems, storage and warehousing limitations, logistics and delivery delays, digital and technology gaps, financial constraints, communication barriers, pricing instability, product quality issues, power supply problems, security concerns, and behavioral practices all affect the chain in different degrees. The mean, standard deviation, and chi-square analysis demonstrated that these challenges are not evenly distributed and are statistically significant. Some constraints are far more dominant than others, especially those tied to logistics, infrastructure, preservation, and institutional support. This means the supply chain is functional, but not efficient in a full sense; it is being held together by demand and production needs while still being weakened by structural constraints.

The fifth objective examined the control standards that regulate the system. This part of the study revealed that the supply chain is also shaped by standards such as government support and policy mechanisms, contractual arrangements, storage and preservation standards, logistics coordination, technological control, cooperative structures, contract compliance, supply chain management practices, supply integration, trust-based relationships, digital tracking, and data-sharing systems. The Likert-scale analysis, weighted means, standard deviations, remarks, and chi-square results showed that these control standards are not merely theoretical. They are active operational mechanisms that influence how products are sourced, transported, monitored, preserved, and coordinated across the farming-hospitality chain. The inferential results confirmed that these standards are statistically significant, meaning the supply system is governed by identifiable organizational and managerial structures rather than by casual or accidental exchange alone.

When all the objectives are read together, a clearer picture emerges. The study shows a supply chain that is real, locally grounded, and functionally active, but also vulnerable to a range of operational weaknesses. Production is strong, hospitality demand is visible, the linkage exists, and control mechanisms are present, but the chain is still constrained by infrastructure, logistics, communication, policy, and coordination problems. That combination is analytically important because it means the issue is not whether the supply chain exists, but how effectively it functions. The findings therefore point to a system with real potential, but one that requires stronger institutional support, better coordination, improved storage and transport systems, clearer contractual structures, more reliable data-sharing, and better digital integration.

In summary, the entire analysis establishes that farming and hospitality industries in Owerri, Imo State are connected through a functional resource supply chain that is statistically identifiable, operationally visible, and economically significant. The chain is supported by local production and strong hospitality demand, but its performance is reduced by a set of major challenges and uneven control standards. The study therefore provides a full picture of the supply-chain environment: what is produced, what is demanded, how the two sectors connect, what disrupts the connection, and what mechanisms exist to regulate it. That makes the work not just descriptive, but diagnostic and solution-oriented.